Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

From 0.756 to 0.846 AUROC

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Background: Breast cancer is the most commonly diagnosed cancer worldwide. Population screening based on full-field digital mammography (FFDM) reduces mortality but imposes heavy radiologist workload and substantial inter-reader variability, motivating computer-aided diagnosis.

Purpose: To develop and evaluate MamNet-6, a six-stage pipeline for breast cancer detection on the VinDr-Mammo FFDM benchmark, progressing from single-view CNN baselines to a multi-view ensemble with uncertainty-aware triage simulation.

Materials and Methods: We trained five single-view CNN backbones (Stage 1), a shared-backbone BreastPairNet multi-view network across twelve backbone×fusion×resolution variants (Stage 2), a CrossView Transformer with spatial cross-attention (Stage 3), and a bounding-box annotation ablation study (Stage 3b). We then applied Monte Carlo (MC) Dropout for uncertainty quantification (Stage 4), assembled a 16-model ensemble with test-time augmentation (TTA) (Stage 5), and evaluated a Verboom-style hybrid triage simulation (Stage 6). External zero-shot validation was performed on the RSNA 2022 dataset from United States screening programmes.

Results: DenseNet-121 attained AUROC=0.7564 (Stage 1). BreastPairNet improved to AUROC=0.8256 (+9.1%, Stage 2); CrossView Transformer reached AUROC=0.8317 (+10.0%, Stage 3). Bounding-box annotations degraded AUROC by 0.12–0.20. The 16-model ensemble achieved AUROC=0.8458 (+11.8%) with sensitivity at 90% specificity (S@90%spec)=0.6566. Hybrid triage at 50% AI reads preserved 87.9% sensitivity under oracle conditions. External RSNA 2022 AUROC=0.6008.

Conclusion: A six-stage multi-view pipeline achieves AUROC=0.8458 on VinDr-Mammo. Bounding-box shortcut learning is a key negative finding; domain generalisation to United States (US) FFDM—as tested on the RSNA 2022 screening dataset—remains an open challenge.

Keywords: breast cancer detection; mammography; MamNet-6; multi-view learning; uncertainty quantification; triage; VinDr-Mammo.

Key Results at a Glance

Sensitivity at 90% specificity (S@90%spec) on VinDr-Mammo: **0.6566** (best ensemble) vs 0.1636 (RSNA 2022 — large domain gap, indicating that the conservative end of the decision surface is most affected by domain shift).

1 Introduction

Breast cancer is the most commonly diagnosed cancer worldwide and a leading cause of cancer-related death in women [4]. Population screening programmes based on full-field digital mammography (FFDM) have repeatedly been shown to reduce breast-cancer mortality, yet they impose a heavy operational burden: in many countries each screening exam is double-read by two radiologists, and reader sensitivity varies by 10–30 percentage points between individuals [10, 11]. False-negative rates of 10–30% and false-positive recall rates of approximately 8% are well documented across both clinical practice and large computer-aided diagnosis literature, with historical hand-engineered detection systems failing to deliver measurable benefit in routine screening [17].

Deep learning has dramatically improved the situation. CNNs trained on hundreds of thousands of exams now reach or exceed individual radiologist AUC on internal and some external cohorts [11], and meta-analyses report pooled AUC of 0.87 for AI on digital mammography versus 0.81 for radiologists [4]. Three trends are particularly relevant. First, multi-view modelling that fuses the craniocaudal (CC) and mediolateral oblique (MLO) projections improves over single-view inference because findings often appear unambiguously in only one view [10]. Second, transformer-based fusion of views allows feature-level interaction rather than score-level averaging. Third, uncertainty-aware inference via Monte Carlo (MC) Dropout [13] enables a *hybrid triage*

Table 1: Summary of the main AUROC results across the six-stage pipeline.

Stage	Method	AUROC	vs. single-view
Prior work	MamT4 [18] (4-view Transformer)‡	0.840	—
Stage 1	DenseNet-121 (best single-view)	0.756	baseline
Stage 2	EfficientNet-B3 +reg (BreastPairNet)	0.826	**+9.1%**
Stage 3	ConvNeXt-Base (CrossView Transformer)	0.832	**+10.0%**
Stage 4	DenseNet-121/concat MC Dropout	0.810	sensitivity ↑
Stage 5	**16-model ensemble + TTA×5**	**0.846**	**+11.8%** ⋆
External	RSNA 2022 (zero-shot transfer)	0.601	domain gap **!**

Input: 20,000 VinDr-Mammo DICOM images from 5,000 patients.
Preprocessing: DICOM decoding, percentile clipping, CLAHE, orientation normalisation.
Stage 1: single-view CNN baselines, best AUROC = 0.7564.
Stage 2: BreastPairNet CC+MLO fusion, best AUROC = 0.8256.
Stage 3: CrossView Transformer, best AUROC = 0.8317.
Stage 3b: annotation ablation, all painted-box variants lose 0.122–0.201 AUROC.
Stage 4: MC Dropout, entropy-ranked uncertainty estimates.
Stage 5: 16-model ensemble with asymmetry features and TTA, best AUROC = 0.8458.
Stage 6: hybrid triage simulation and external RSNA 2022 validation.

Figure 1: MamNet-6: overview of the six-stage breast cancer detection pipeline.

protocol in which the AI reads the most-confident exams autonomously and refers the rest to a human reader [12], reducing radiologist workload by approximately 38% in retrospective evaluation [12].

The VinDr-Mammo benchmark [1] is a 20,000-image, patient-level, 4-view FFDM dataset. Three gaps in the literature are evident. **(i) Single-view dominance.** Most published VinDr-Mammo results train one image at a time. **(ii) Spatial cross-view attention is underexplored.** Existing multi-view designs typically concatenate global features. **(iii) Uncertainty quantification and triage simulation are rare.** No published study has applied MC Dropout to VinDr-Mammo and simulated a Verboom-style triage policy [12].

This paper makes seven contributions. First, we present a systematic comparison of five single-view CNN backbones. Second, we introduce **BreastPairNet**, a shared-backbone multi-view network evaluated across twelve backbone×fusion×resolution variants. Third, we propose a **CrossView Transformer** with bidirectional spatial cross-attention between views. Fourth, a controlled **bounding-box annotation ablation** documents a consistent negative result across all four architecture pairs. Fifth, a **16-model ensemble** with TTA achieves AUROC=0.8458, the best result in this study. Sixth, **MC Dropout** provides calibrated uncertainty estimates with a statistically meaningful entropy gap between positive and negative cases. Seventh, a **hybrid triage simulation** based on the Verboom framework demonstrates 50% workload reduction while preserving 87.9% sensitivity under oracle conditions.

Figure 1 illustrates the full pipeline.

2 Related Work

Deep CNN backbones. Residual networks [7] and densely connected networks [8] underpin almost all modern screening AI systems. CheXNet [6] established DenseNet-121 as a default starting point for chest imaging. EfficientNet [9] introduced compound scaling (B3: 12M parameters). ConvNeXt [16] modernised pure-convolutional design to transformer-level ImageNet accuracy.

Multi-view mammography. Wu et al. [10] trained a four-stream ResNet-22 on 229,426 NYU exams (AUC=0.895), establishing that shared backbones and feature-level aggregation are essential under limited data. McKinney et al. [11] reported AUC=0.889 on UK FFDM, outperforming six radiologists. Cross-view spatial attention outperforms score-level averaging, motivating Stage 3.

VinDr-Mammo-specific work. Nguyen et al. [1] released VinDr-Mammo with BI-RADS, density, and lesion-level annotations. Ibragimov et al. [18] proposed MamT4 (AUC=0.840±0.017). Abdikenov et al. [2] demonstrated CLAHE+YOLOv12 mAP gains; Mellado et al. [3] showed competitive patch-level F1 under limited annotation.

Class imbalance and focal loss. Focal loss [5] suppresses easy negatives via $(1-p)^\gamma$ down-weighting; Johnson & Khoshgoftaar [15] survey algorithm-level imbalance methods.

Uncertainty quantification. Kendall and Gal [13] formalised aleatoric and epistemic uncertainty; MC Dropout provides tractable Bayesian model averaging. Verboom et al. [12] demonstrated 38.1% workload reduction on 41,469 Dutch screening exams while preserving cancer detection rate.

3 Dataset

3.1 VinDr-Mammo (primary dataset)

VinDr-Mammo [1] contains 20,000 FFDM images from 5,000 patients, four views per study (R-CC, L-CC, R-MLO, L-MLO) stored as DICOM, with BI-RADS (1–5), density (A–D), and lesion-level bounding box annotations (Table 2).

Table 2: BI-RADS distribution in VinDr-Mammo and binary label mapping.

BI-RADS	Images	Binary label
1	14,036	negative
2	4,144	negative
3	832	negative
4	824	**positive**
5	164	**positive**
Total	**20,000**	988 positive (4.94%)

The official split is 4,000 training patients (16,000 images) and 1,000 test patients (4,000 images). For multi-view experiments we construct one ipsilateral CC+MLO pair per breast, yielding 8,000 training pairs and 2,000 test pairs, of which 99 test pairs (4.95%) are cancer-positive. A 5% validation slice (~400 pairs) is held out for early stopping; the official test split is never used for model selection.

3.2 RSNA 2022 (external dataset)

The RSNA 2022 dataset comprises 54,706 screening DICOMs from 11,913 patients. After 8-bit normalisation identical to VinDr-Mammo preprocessing, approximately 29,519 images required `pylibjpeg` to decode JPEG Lossless DICOMs. Non-CC/MLO views were excluded. The final paired dataset contains 23,826 CC+MLO pairs (19,062 training, 4,764 validation) on an 80/20 patient-level split.

4 Methods

4.1 DICOM preprocessing (12/14-bit → 8-bit PNG)

Four-step pipeline: (1) decode with pydicom+`pylibjpeg`; (2) percentile-clip to [1%, 99%]; (3) CLAHE [14] (32×32 tiles, clip-limit 2.0); (4) standardise breast orientation so that the breast faces right—a horizontal flip is applied when the mean intensity of the left image half exceeds the right, since breast tissue is brighter than the background field. Output: single-channel 8-bit PNG replicated to 3 channels.

4.2 Single-view CNN backbones

Five configurations: **VGG19**, **ResNet-50** (224 px), **ResNet-50** (384 px), **EfficientNet-B3** (224 px), **DenseNet-121** (224 px). All use ImageNet weights, focal loss (α=0.25, γ=2.0) [5], WeightedRandomSampler, and AdamW with cosine decay. Training: 20 epochs with early stopping on validation AUROC.

4.3 BreastPairNet (Stage 2)

Shared backbone produces per-view global-average-pooled vectors $\mathbf{f}_{CC}, \mathbf{f}_{MLO} \in \mathbb{R}^d$. Two fusion heads: **concatenation** (2d-dim MLP, dropout 0.5) and **attention** (learned per-view scalar weights via softmax). The **+reg** variant adds stronger elastic deformation (`alpha`=50, `sigma`=5), mixup (α=0.2), and label smoothing (0.1). Training: backbone frozen 3 epochs, then uniform learning rate 1e-4 applied to all parameters. Figure 2 illustrates the architecture.

4.4 CrossView Transformer (Stage 3)

Replaces global-pooled fusion with spatial cross-attention. Both views pass through the shared CNN backbone to the last spatial feature map; spatial tensors are flattened to tokens and entered into a two-layer Transformer encoder where queries from each view attend to keys/values of the other. Cross-attended outputs are global-averaged and concatenated for the final classifier. Figure 3 contrasts this with BreastPairNet.

4.5 Bounding-box annotation (training only)

Ground-truth boxes (VinDr-Mammo) were drawn as white outlines (40-px line width) on training images only. For RSNA 2022, a YOLOv8s detector pre-trained on VinDr-Mammo was used (0.8% sensitivity — noisy hints only). Test sets always used original unannotated PNGs to prevent label leakage.

4.6 MC Dropout (Stage 4)

MC Dropout was chosen for uncertainty quantification because it requires no architectural modifications: dropout layers that are normally disabled at test time are kept active, converting a single point prediction into a distribution over outputs without retraining [13]. This makes entropy-based case ranking possible at negligible computational cost.

Concretely, BatchNorm is kept in eval mode while dropout remains active. $N = 20$ stochastic forward passes per test pair yield probabilities $\{p_i\}_{i=1}^{20}$; their mean $\bar{p}$ is used for classification and the binary entropy $H(\bar{p}) = -\bar{p}\log\bar{p} - (1-\bar{p})\log(1-\bar{p})$ quantifies predictive uncertainty. The classification threshold is selected by maximising Youden's J index (J = sensitivity + specificity − 1) on the validation set.

4.7 Hybrid triage (Verboom 2025 framework)

Following Verboom et al. [12], cases are sorted by ascending predictive entropy; the K most-confident fraction is routed to AI and the remainder to a radiologist (modelled as oracle or 85% sensitivity). Overall sensitivity = ($\mathrm{TP}_{\mathrm{AI}}$ + cancers referred) / total cancers.

4.8 Training and evaluation

All models trained on NVIDIA RTX 5070 (12 GB VRAM) with FP16 mixed-precision (AMP) and AdamW (weight decay 1e-4 for Stage 1; 1e-3 for Stage 2). Primary metric: AUROC. Also reported: F1 at Youden threshold, sensitivity, specificity, and **S@90%spec** (sensitivity at 90% specificity). 95% percentile bootstrap confidence intervals (n=2000 resamples) on the 2,000-pair test split (99 cancer-positive). Typical AUROC CI

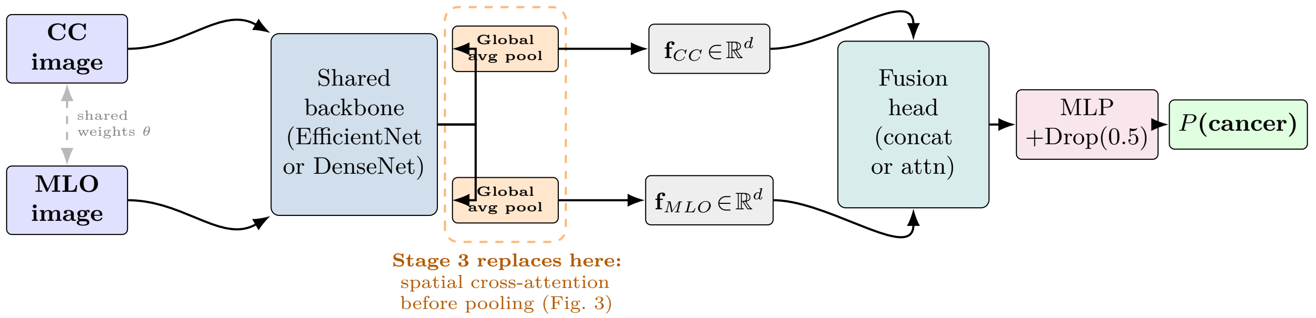

Figure 2: BreastPairNet architecture (Stage 2 of MamNet-6). CC and MLO images from the same breast pass through a *shared* CNN backbone (weights θ tied for both views). Each view is independently reduced to a global-average-pooled (GAP) feature vector $\mathbf{f} \in \mathbb{R}^d$; the two vectors are fused by concatenation or attention, then classified by a two-layer MLP with 50% dropout. The orange dashed box highlights the GAP step — the key design decision that Stage 3 (CrossView Transformer, Figure 3) replaces with bidirectional spatial cross-attention applied *before* pooling.

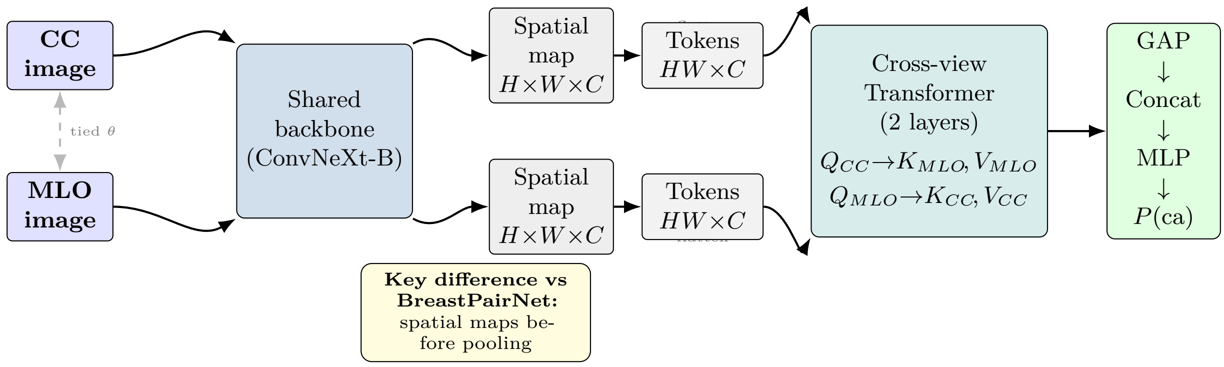

Figure 3: CrossView Transformer architecture (Stage 3). Unlike BreastPairNet, the backbone output is *not* global-averaged before fusion. Spatial feature maps are flattened to token sequences; a two-layer Transformer encoder applies bidirectional cross-attention (Q_{CC} attends to K, V from MLO and vice versa), enabling the model to localise spatially consistent findings across views before pooling. Best: ConvNeXt-Base, AUROC=0.8317.

Table 3: Single-view CNN baselines on VinDr-Mammo (n=16,000 train, n=4,000 test images).

Architecture	Image size	Sensitivity	Specificity	F1	AUROC	S@90%
VGG19	224 px	0.2626	0.9835	0.3323	0.7066	0.3990
ResNet-50	224 px	0.2677	0.9847	0.3430	0.7286	0.4495
EfficientNet-B3	224 px	0.2980	0.9687	0.3138	0.7330	0.3990
ResNet-50 (384 px)	384 px	0.2879	0.9821	0.3529	0.7489	0.4495
DenseNet-121	**224 px**	**0.2475**	**0.9882**	**0.3356**	**0.7564**	**0.4444**

width: ±0.04–0.05.

5 Experiments and Results

5.1 Stage 1 — Single-view CNN baselines

DenseNet-121 achieves the best AUROC (0.7564) but only 24.8% sensitivity at the default threshold (Table 3; Figure 4). All five models overfit (train F1 ≈ 0.98) and reach only 40–45% sensitivity at the 90%-specificity operating point — well below the ∼85% radiologist sensitivity in screening literature [17].

5.2 Stage 2 — Multi-view BreastPairNet

The best single model is **EfficientNet-B3 + attention + +reg** at AUROC=0.8256 (95% CI 0.7755–0.8707), a +9.1% relative gain over DenseNet-121 single-view (Table 4). Each row in Table 4 reports the best-checkpoint result from a single training run evaluated on the held-out test split. Standard deviations across runs are not reported as each configuration was trained once; the 95% percentile bootstrap CIs quantify test-split sampling variability, not run-to-run variability.

All point estimates in Table 4 and Table 6 are obtained by re-evaluating each best-checkpoint model on the held-out test split (n=2,000 pairs, 99 positives) at the Youden-optimal threshold, with 95% percentile bootstrap confidence intervals (n=2,000 resamples). Validation AUROC curves for the best Stage 2 and Stage 3 models are shown in Figure 6.

Three observations stand out (Table 5; Figure 5). **(i) +reg consistently lifts AUROC** by from 0.008 to 0.021 across all backbone/fusion pairs. **(ii) Resolution scaling is not uniformly beneficial**: DenseNet-121

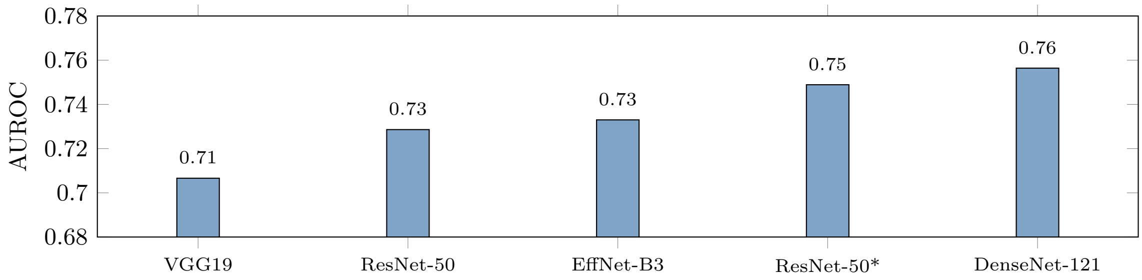

Figure 4: Stage 1: AUROC by single-view architecture. ResNet-50* uses 384 px; all others 224 px.

Table 4: BreastPairNet ablation across backbone, fusion, and resolution.

Architecture	Fusion	Image	Best AUROC [95% CI]	F1	Sens	S@90% [95% CI]
ResNet-50	concat	384 px	0.7467 [0.6921–0.7998]	0.2350	0.5354	0.4646 [0.3736–0.5673]
ResNet-50	attention	224 px	0.7847 [0.7324–0.8338]	0.2522	0.5859	0.5051 [0.4023–0.6019]
ResNet-50	attention	512 px	0.7922 [0.7339–0.8429]	0.2093	0.7475	0.5152 [0.4130–0.6214]
ResNet-50	concat	224 px	0.7962 [0.7428–0.8445]	0.2363	0.6970	0.4949 [0.3958–0.5977]
DenseNet-121	attention	224 px	0.8014 [0.7487–0.8515]	0.2800	0.6364	0.5758 [0.4660–0.6667]
EfficientNet-B3	attention	224 px	0.8048 [0.7530–0.8556]	0.2138	0.7677	0.5455 [0.4476–0.6449]
DenseNet-121	concat	224 px	0.8073 [0.7577–0.8504]	0.2101	0.7374	0.5152 [0.4141–0.6100]
DenseNet-121 **+reg**	attention	224 px	0.8097 [0.7594–0.8591]	0.2033	0.7980	0.5556 [0.4301–0.6458]
DenseNet-121	attention	512 px	0.8150 [0.7675–0.8617]	0.1928	0.8081	0.5455 [0.4476–0.6408]
DenseNet-121 **+reg**	concat	224 px	0.8177 [0.7659–0.8649]	0.2357	0.7677	0.6061 [0.5047–0.7000]
EfficientNet-B3	attention	512 px	0.8175 [0.7698–0.8661]	0.2807	0.7273	0.5556 [0.4495–0.6505]
EfficientNet-B3 +reg	**attention**	**224 px**	**0.8256 [0.7755–0.8707]**	**0.3701**	**0.6263**	**0.6263 [0.5229–0.7143]**

Table 5: Isolated effect of the +reg training recipe.

Backbone	Fusion	Base AUROC	+reg AUROC	Gain
DenseNet-121	concat	0.8073	0.8177	+0.010
DenseNet-121	attention	0.8014	0.8097	+0.008
EfficientNet-B3	attention	0.8048	**0.8256**	**+0.021**

benefits from 512 px (+0.014), but EfficientNet-B3 +reg at 224 px (0.8256) beats 512 px without +reg (0.8175). **(iii) ResNet-50 degrades at 384 px** (0.7467 vs 0.7962 at 224 px) due to batch-size reduction under VRAM pressure.

5.3 Stage 3 — CrossView Transformer

5.3.1 Performance without annotation

ConvNeXt-Base achieves the best AUROC at 0.8317 (+10.0% over single-view DenseNet-121). DenseNet-121 with VinDr-Mammo+RSNA 2022 joint training reaches AUROC=0.8028, confirming incremental cross-dataset benefit. Chest-X-ray pretraining via the torchxrayvision library[1] underperforms ImageNet initialisation (AUROC 0.7648 vs 0.7874), consistent with domain mismatch between lung radiographs and mammographic texture.

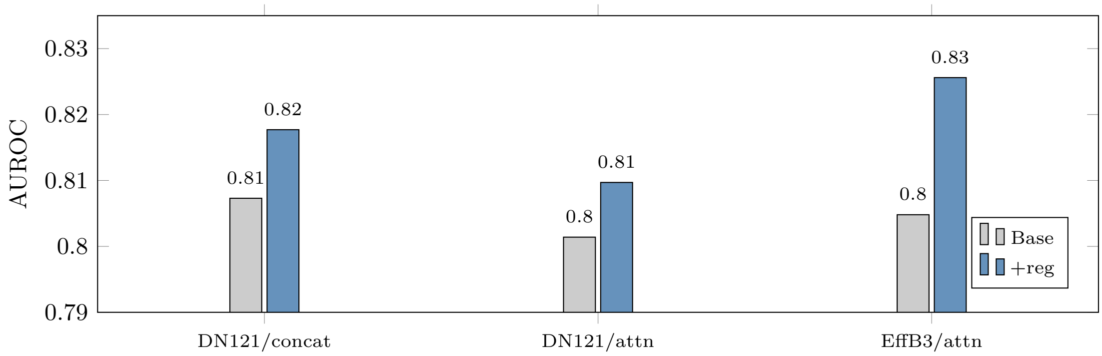

Figure 5: Stage 2: Effect of +reg on AUROC for three backbone/fusion pairs.

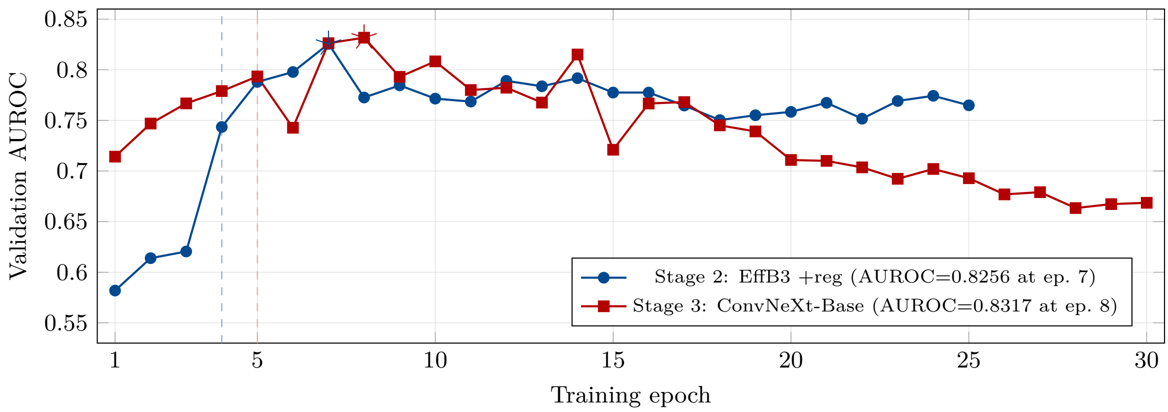

Figure 6: Validation AUROC vs. training epoch for the best Stage 2 (EffB3 +reg) and Stage 3 (ConvNeXt-Base) models. Vertical dashed lines mark the backbone unfreeze point (epoch 4 for EffB3, epoch 5 for ConvNeXt). Stars ($\star$) indicate the best-checkpoint epoch used for test-set evaluation. Both models overfit after their best epoch as training loss continues to decrease while validation AUROC plateaus.

Table 6: CrossView Transformer results without bounding-box annotation.

Architecture	Training data	Best AUROC [95% CI]	F1	Sens	S@90% [95% CI]
DenseNet-121 (medical init)	VinDr	0.7648 [0.7116–0.8210]	0.2374	0.5960	0.5051 [0.4020–0.6100]
EfficientNet-B3	VinDr	0.7795 [0.7276–0.8300]	0.2022	0.7273	0.4545 [0.3474–0.5500]
DenseNet-121	VinDr	0.7874 [0.7299–0.8401]	0.2678	0.6465	0.5556 [0.4474–0.6482]
DenseNet-121	VinDr+RSNA	0.8028 [0.7484–0.8536]	0.2640	0.6667	0.5354 [0.4433–0.6404]
ConvNeXt-Base	**VinDr**	**0.8317 [0.7832–0.8753]**	**0.2587**	**0.7475**	**0.5758 [0.4737–0.6768]**

Table 7: Bounding-box annotation ablation. Test set always uses original unannotated PNGs.

Architecture	No annotation AUROC [95% CI]	With annotation AUROC [95% CI]	ΔAUROC
DenseNet-121	0.7874 [0.7299–0.8401]	0.5862 [0.5274–0.6433]	-0.201
DenseNet-121 +RSNA	0.8028 [0.7484–0.8536]	0.6675	-0.135
DenseNet-121 (medical)	0.7648 [0.7116–0.8210]	0.6433 [0.5886–0.6966]	-0.122
EfficientNet-B3	0.7795 [0.7276–0.8300]	0.6020 [0.5460–0.6617]	-0.178

5.3.2 Bounding-box annotation ablation

All four pairs show annotation consistently degrades AUROC by 0.122–0.201 (Table 7; Figure 7). At epoch 4 the backbone unfreezes and fits the white box outlines as a shortcut texture; training F1 reaches ~0.97 while test AUROC collapses because boxes are absent at test time. Potential remedies may include: (a) box-dropout regularisation, in which annotation boxes are randomly masked during training to prevent over-reliance on the shortcut; (b) test-time box injection via a lesion detector, so the test distribution matches training; or (c) cropping the input to

[1] torchxrayvision: `github.com/mlmed/torchxrayvision`

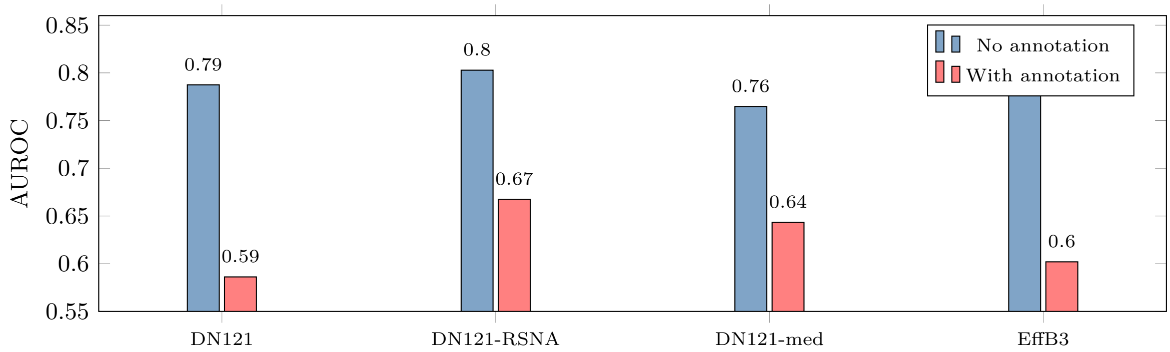

Figure 7: Stage 3b: Annotation consistently degrades AUROC due to shortcut learning and test-time distribution shift.

the bounding-box region, passing only the annotated content to the model rather than the full image with a drawn outline. These suggestions are speculative; we are not aware of published evidence validating them specifically for mammography annotation shortcuts.

5.4 Stage 4 — MC Dropout uncertainty quantification

Table 8: MC Dropout vs deterministic inference (DenseNet-121/concat).

Mode	AUROC	Sens	Spec	S@90%	Thr.
Deterministic	0.8073	0.7374	0.7249	0.5152	0.3304
MC Dropout	**0.8099**	**0.7677**	**0.7007**	**0.5253**	**0.3138**

MC Dropout slightly improves AUROC (+0.003) and produces a meaningful sensitivity shift (+3.0 percentage points, Table 8; inference flow in Figure 8). More importantly, the mean entropy is **0.5670** on cancer-positive pairs versus **0.5112** on negatives (Table 9) — a 0.056-nat gap that enables the Stage 6 triage policy.

Table 9: Predictive entropy by class ($N = 20$ MC passes, 2,000 test pairs).

Entropy	Negatives ($n = 1{,}901$)	Positives ($n = 99$)
Mean	0.5112	0.5670
Median	0.5310	0.6221
Std dev	0.1436	0.1370
Gap: +0.056 nats. Higher entropy $\Rightarrow$ refer high-uncertainty cases.		

5.5 Stage 5 — Model ensemble

5.5.1 Original ensemble (16 BreastPairNet models)

Mean-probability ensemble of 16 BreastPairNet checkpoints combined with three breast-asymmetry models and five-fold test-time augmentation (TTA×5). **Best result in the study**: AUROC=0.8458, sensitivity=0.7374, specificity=0.8296, F1=0.2944, S@90%=0.6566, S@95%=0.5657 at threshold 0.3414 (TP=73, TN=1577, FP=324, FN=26; Table 10).

The per-class breakdown is shown in Table 11. This ensemble does **not** include ConvNeXt-Base or EfficientNet-B3 +reg models.

5.5.2 Updated ensemble (25 checkpoints)

Adding all CrossView checkpoints (including lower-epoch snapshots) gives AUROC=0.8441, S@90%=0.6566 (unchanged), sensitivity=0.7677. The marginal AUROC drop confirms that weaker models drag the mean under mean combination; S@90% is robust.

5.5.3 Selective ensemble (top-3 +reg + top-5 CrossView + 3 asymmetry)

11 models, TTA×5: AUROC=0.8446, S@90%=0.6667 (+0.010 over baseline). Marginally better at the clinical operating point; difference is within sampling noise.

5.5.4 Ensemble comparison

Table 12 summarises all three configurations. Full metrics (specificity, F1, threshold) were only computed for the primary 16-model configuration; secondary configurations were evaluated on AUROC and the clinical operating point (S@90%) only.

5.6 Stage 6 — Hybrid triage simulation

Under oracle assumptions, **K=50% delivers 87.9% overall sensitivity with 50% workload reduction** (Table 13; Figure 9). Under realistic 85% radiologist sensitivity [17], the same K=50% yields 77.3% — below the clinical floor. Achieving ≥80% overall sensitivity under realistic conditions requires K≤40% (Table 14).

5.7 Comparison to literature

AUROC progression across the pipeline is summarised in Table 15 and visualised in Figure 10. The ‡ symbol in Table 15 flags a methodological difference: MamT4 [18] excludes BI-RADS 3 cases from evaluation, while our pipeline retains them as negative, making the AUROC values not directly comparable. McKinney et al. [11] (AUC=0.889) and Yoon et al. [4] (pooled AUC=0.87) report on different datasets and are likewise not directly comparable.

5.8 External validation on RSNA 2022

The best ensemble (AUROC=0.8458 on VinDr-Mammo) was applied zero-shot to RSNA 2022. Full

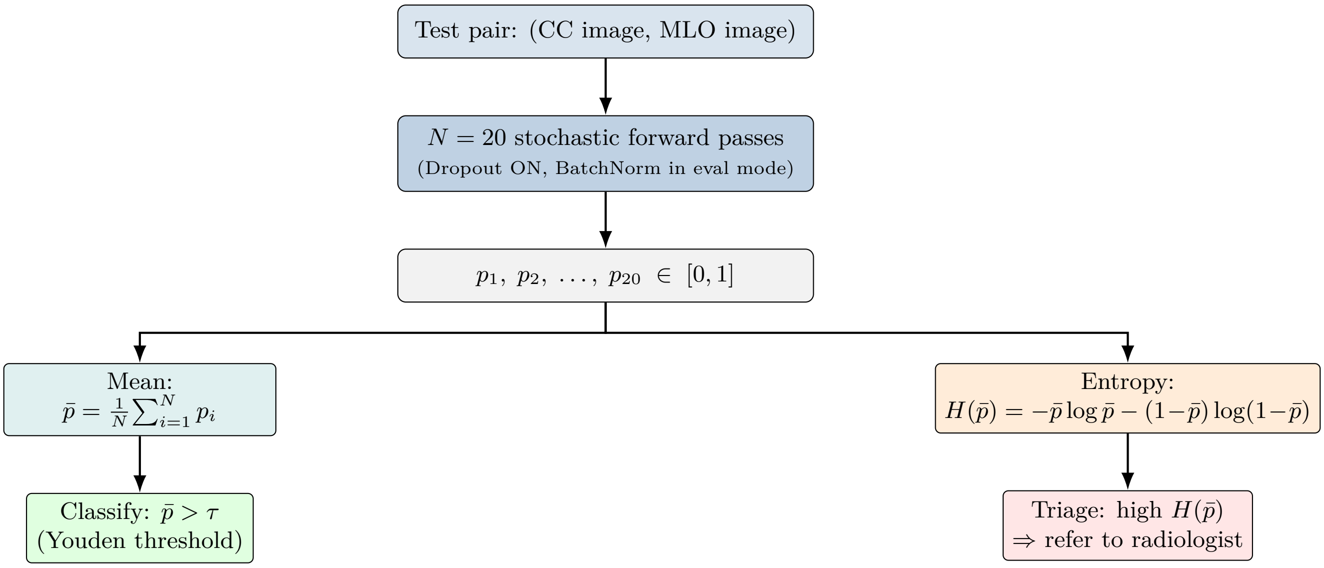

Figure 8: MC Dropout inference (Stage 4). Dropout layers remain active at test time while BatchNorm uses running statistics. Twenty stochastic passes produce a probability distribution $\{p_i\}_{i=1}^{20}$. The mean $\bar{p}$ is used for classification at the Youden-optimal threshold τ; the binary entropy $H(\bar{p})$ ranks cases by epistemic uncertainty and routes high-uncertainty cases to a radiologist in the Stage 6 triage protocol.

Table 10: Best ensemble result (mean probability, TTA×5).

Method	AUROC	Sens	Spec	F1	S@90%	S@95%	Threshold
Ensemble (N=16, TTA=5)	**0.8458**	**0.7374**	**0.8296**	**0.2944**	**0.6566**	**0.5657**	**0.3414**

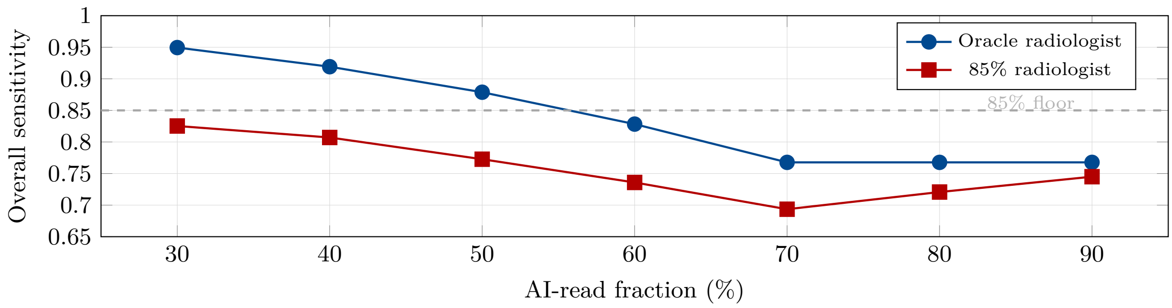

Figure 9: Stage 6: Overall sensitivity vs. AI-read fraction. Dashed line = 85% clinical sensitivity floor.

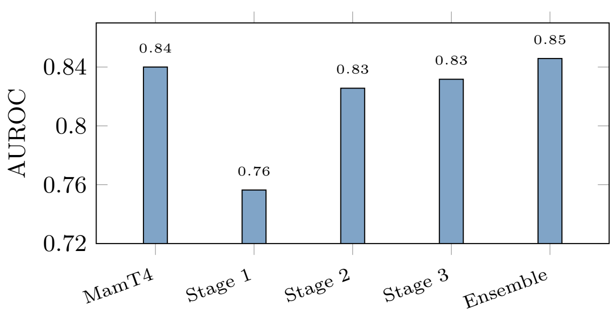

Figure 10: AUROC progression across pipeline stages vs. MamT4 prior work.

Table 11: Confusion matrix at Youden threshold ($t = 0.3414$).

	Pred. negative	Pred. positive
Actual negative	TN = 1,577	FP = 324
Actual positive	FN = 26	TP = 73

metrics are shown in Table 16; the per-metric performance drop is detailed in Table 17.

Three factors may drive the gap: (1) Vietnamese vs United States breast density distribution and lesion prevalence [1, 11]; (2) different FFDM equipment and post-processing pipelines; (3) the S@90%spec drop (−75.1%) is particularly stark at the clinical operating point, suggesting the conservative side of the decision surface is most affected by domain shift. Table 18 sum-

Table 12: Comparison of ensemble configurations (TTA×5, mean-probability combination).

Configuration	Models	AUROC	Sens	S@90%
Original (BreastPairNet)	16	**0.8458**	0.7374	0.6566
Updated (+CrossView)	25	0.8441	0.7677	0.6566
Selective (+reg+CVT+asym)	11	0.8446	—	**0.6667**

—: not evaluated. Differences in AUROC are within the ±0.04 percentile bootstrap CI; the selective ensemble offers marginal S@90% benefit.

Table 13: Hybrid triage sweep (oracle vs. 85% radiologist sensitivity).

Frac	Wkld Δ	Sen (or.)	Sen (85%r)	AI Sen	Recall
30%	-30%	0.9495	0.8253	0.7059	0.0017
40%	-40%	0.9192	0.8071	0.6800	0.0013
50%	**-50%**	**0.8788**	**0.7727**	**0.5862**	**0.0031**
60%	-60%	0.8283	0.7359	0.5526	0.0052
70%	-70%	0.7677	0.6935	0.5400	0.0133
80%	-80%	0.7677	0.7207	0.6618	0.1305
90%	-90%	0.7677	0.7450	0.7262	0.2238

Table 14: Max AI-read fraction by sensitivity floor.

Floor	Oracle	85% rad	Workload Δ
≥90%	40%	n/a	-40%
≥85%	50–60%	n/a	-50% to -60%
≥80%	60%	40%	-40% to -60%
≥77%	70%	50%	-50% to -70%

Table 15: AUROC progression on VinDr-Mammo binary cancer detection.

Method	AUROC
MamT4 [18]‡	0.840
Ours Stage 1 (DenseNet-121)	0.7564
Ours Stage 2 (BreastPairNet)	0.8256
Ours Stage 3 (CrossView)	0.8317
Ours Stage 5 (Ensemble)	**0.8458**

Table 16: External validation: VinDr-Mammo (2,000 pairs, 99 cancers) vs. RSNA 2022 (4,764 pairs, 110 cancers).

Dataset	AUROC [95% CI]	S@90%	Sens	Spec
VinDr-Mammo	0.8458	0.6566	0.7374	0.8296
RSNA 2022	0.6008 [0.5577–0.6585]	0.1636	0.3909	0.7598

Table 17: Performance drop: VinDr-Mammo → RSNA 2022.

Metric	VinDr-Mammo	RSNA 2022	Drop	Rel.
AUROC	0.8458	0.6008	-0.245	-29%
S@90%spec	0.6566	0.1636	**-0.493**	**-75%**
Sensitivity	0.7374	0.3909	-0.347	-47%
Specificity	0.8296	0.7598	-0.070	-8%

marises the best-performing model family at each stage.

Table 18: Summary of the best-performing model family in each stage.

Family	Best AUROC	95% CI	S@90%	Key advantage
Single-view CNN	0.7564	—	0.4444	Simple baseline
BreastPairNet (concat)	0.8177	0.7659–0.8649	0.6061	Robust across backbones
BreastPairNet (attention)	0.8256	0.7755–0.8707	0.6263	Best regularised
CrossView Transformer	0.8317	0.7832–0.8753	0.5758	Spatial alignment
Ensemble (16+3, TTA×5)	**0.8458**	—	**0.6566**	Best overall ⋆

6 Discussion

Multi-view fusion. The +9.1% AUROC gain from BreastPairNet is consistent with the radiological observation that lesions may appear unambiguously in only one mammographic projection. Joint conditioning of CC and MLO views likely reduces view-specific false positives, though the individual contributions of shared weights versus the fusion architecture cannot be isolated from these experiments alone.

Regularisation benefit. With only ∼395 cancer-positive training pairs, the limited positive sample count may explain why stronger augmentation (elastic deformation, mixup, label smoothing) acts as a powerful implicit regulariser. AUROC gains from 0.008 to 0.021 were observed consistently across all backbone/fusion pairs tested.

CrossView Transformer vs BreastPairNet. ConvNeXt-Base (AUROC=0.8317) vs EfficientNet-B3 +reg (AUROC=0.8256): Δ=0.006, which is within the ±0.04 percentile bootstrap CI—these models are not statistically separated. One plausible explanation is that exchanging spatial feature maps before global pooling may preserve localisation information that BreastPairNet's independent pooling discards, though confounded by the different backbone architectures.

Medical pretraining. Chest-X-ray features (lung parenchyma, mediastinum, rib edges) may be domain-mismatched to mammographic texture, which could explain why ImageNet initialisation outperformed torchxrayvision chest-X-ray weights (AUROC 0.7874 vs 0.7648). However, as these configurations also differ in other ways, this interpretation is tentative.

Bounding-box shortcut learning. Painted boxes provide a texture shortcut perfectly correlated with the cancer label during training but absent at test time. This pattern is consistent with well-documented shortcut learning phenomena in medical imaging. Potential remedies may include box-dropout regularisation or test-time box injection via a lesion detector.

MC Dropout uncertainty. The +0.003 AUROC gain from MC Dropout is within the 95% percentile bootstrap CI and should not be over-interpreted. The more clinically meaningful finding is the sensitivity boost (+3.0 percentage points, AUROC 0.8073→0.8099) and the 0.056-nat entropy gap between cancer-positive and cancer-negative classes, which enables the Stage 6 triage protocol.

Triage simulation. Under realistic 85% radiologist sensitivity [17], K=50% yields only 77.3% overall sensitivity—10.6 percentage points below the oracle result (87.9%). These simulations suggest K≤40% may be required to maintain overall sensitivity above the 80% clinical floor; prospective validation with real reader sensitivity data would be needed to confirm these thresholds.

Confidence intervals. With only 99 cancer-positive test pairs, percentile bootstrap CI width is ±0.04–0.05 AUROC. Differences of less than 0.01 AUROC should be treated as within sampling noise. BreastPairNet (AUROC=0.8256, 95% CI 0.7755–0.8707) and CrossView Transformer (AUROC=0.8317, 95% CI 0.7832–0.8753) are not statistically separated.

Limitations. (i) Small positive test count (99 pairs) limits fine-grained comparison; (ii) large domain gap to United States FFDM (AUROC=0.6008 on RSNA 2022); (iii) MamT4 comparison uses a slightly different BI-RADS exclusion; (iv) 7,054 non-CC/MLO RSNA images excluded from pair construction.

Future work. (1) Box-dropout or test-time box injection; (2) CrossView at 512 px with gradient checkpointing; (3) Validate on CBIS-DDSM and INbreast; (4) Tighter ensemble of +reg BreastPairNet + best CrossView + asymmetry; (5) Fine-tune on RSNA training pairs; (6) Prospective triage study with real reader sensitivity data.

7 Conclusion

We have presented MamNet-6, a six-stage pipeline for breast cancer detection on VinDr-Mammo: **Stage 1** (DenseNet-121, AUROC=0.7564) → **Stage 2** (BreastPairNet +reg, AUROC=0.8256, +9.1%) → **Stage 3** (CrossView Transformer, AUROC=0.8317, +10.0%) → **Stage 4** (MC Dropout uncertainty quantification, entropy gap=0.056 nats) → **Stage 5** (16-model ensemble with TTA, AUROC=0.8458, +11.8%) → **Stage 6** (hybrid triage simulation). Bounding-box annotations consistently degraded AUROC by 0.12–0.20 due to shortcut learning—the most actionable negative finding. MC Dropout produced a 0.056-nat entropy gap enabling hybrid triage: K=50% AI reads preserves 87.9% sensitivity under oracle conditions, but only 77.3% under realistic 85% radiologist sensitivity, suggesting K≤40% is required to stay above the 80% clinical floor. External validation (RSNA 2022 AUROC=0.6008) reveals that domain generalisation to United States (US) FFDM—as tested on the RSNA 2022 screening dataset—remains a critical open challenge.

Author Contributions

Haoming Luo: Software, Formal analysis, Investigation, Visualization, Writing – original draft.

Matthew Hamilton: Conceptualization, Methodology, Supervision, Writing – review & editing.
Oscar Meruvia-Pastor: Supervision, Writing – review & editing.
Edward Kendall: Supervision, Writing – review & editing.

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